Engineering Research Report: Comprehensive Design, Sizing, and Vulnerability Analysis of the "Good Boy Limiter" Architecture

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May 5, 2026

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1 Abstract and System Overview

The design of high-power DC distribution networks frequently encounters the engineering challenge of managing massive capacitive loads during the initial charging phase. The "Good Boy Limiter" (GBL) represents an hybrid power management and in-rush current control architecture specifically designed to address these challenges within a 48 V DC ecosystem.

The main problem addressed by the GBL architecture is the safe charge and continuous operation of a downstream load characterized by a really large input capacitance. Specifically, the system must interface with a bulk bus capacitance (C_{tot}) of 7920 μF, which is physically realized through a distributed 4×6 matrix of 330 μF electrolytic capacitors. Furthermore, the system is designed to support a substantial nominal operating current of 260 A, although it is currently constrained to a 120 A limit for safety reasons.

Attempting to directly connect a 48 V stiff voltage source to an uncharged 7920 μF capacitor bank results in an instantaneous "in-rush" current theoretically limited only by the Equivalent Series Resistance (ESR) of the capacitors and the parasitic inductance of the interconnecting cables. In practical terms, this transient current can easily exceed several thousands of amperes. Such catastrophic current spikes induce severe electromagnetic interference (EMI), trigger upstream power supply overcurrent protections, and instantly weld the contacts of mechanical relays due to localized plasma arcing.

To resolve these conflicting requirements, the GBL employs a mixed relay/solid-state architecture. A heavy-duty mechanical external relay (actuated via a 12 V coil) serves as the primary conduction path during nominal steady-state operation. In parallel, a highly controlled solid-state pre-charge circuit, orchestrated by the Texas Instruments LM5069-2 hot-swap controller and an STMicroelectronics STB15810 MOSFET, governs the initial energization phase.

This report provides an exhaustive analysis of the pre-charge dynamics, focusing on the mathematical sizing of the LM5069-2 control parameters. It critically examines the thermal paradigm shift achieved by the inclusion of a series limiting resistor, assesses the Safe Operating Area (SOA) of the selected MOSFET, and documents the role of the ESP32-S3 microcontroller in mitigating topological safety vulnerabilities.

2 Power Architecture: Relay Versus Solid-State

High-current DC power distribution systems inherently suffer from the physical limitations of their switching elements. The architectural decision to utilize a hybrid system in the Good Boy Limiter is grounded in fundamental power dissipation physics.

2.1 The Limitations of Pure Solid-State Switching

In a hypothetical pure solid-state architecture managing the full 260 A nominal current, the conduction losses would be to high. Even utilizing state-of-the-art power MOSFETs exhibiting an ultra-low drain-to-source on-resistance ($R_{DS(on)}$) of 1 mΩ, the continuous power dissipation (P_{loss}) would be substantial. According to Joule's law:

$$P_{loss} = I^2 \times R_{DS(on)} = (260\ \text{A})^2 \times 0.001\ \Omega = 67.6\ \text{W} \tag{1}$$

Dissipating 67.6 W of pure thermal energy on a printed circuit board (PCB) requires massive copper planes, extensive thermal via arrays, large extruded aluminum heatsinks, and mandatory active forced-air cooling. This increases volumetric footprint, acoustic noise, and overall system failure rates.

2.2 The Limitations of Pure Mechanical Switching

Vice-versa, a purely mechanical relay architecture offers micro-ohm contact resistance, virtually eliminating steady-state thermal dissipation. However, mechanical relays are physically incapable of closing into an uncharged 7920 μF capacitive bus at 48 V. As the relay contacts physically approach one another during closure, the dielectric breakdown voltage of the microscopic air gap is exceeded. This initiates a high-energy plasma arc. The thousands of amperes of in-rush current flowing through this plasma instantly melt the contact plating, leading to contact welding and permanent failure of the relay.

2.3 The Hybrid Resolution

The GBL architecture resolves this dichotomy via a parallel power path topology:

- **The Solid-State Pre-Charge Path:** Engaged first, this path utilizes a series-pass N-channel MOSFET (STMicroelectronics STB15810) and a 50 Ω wirewound power resistor (EWWR0010J50R0T9) to strictly limit the transient charging current. The capacitors are charged safely over a calculated time domain.
- **The Main Relay Path:** Engaged only when the ESP32-S3 microcontroller verifies that the bus voltage has reached the nominal source voltage. Because the voltage differential (ΔV) across the relay contacts is near zero, the corresponding current during closure is zero. Closing the relay under this Zero-Voltage Switching (ZVS) entirely eliminates arcing, preserving contact longevity. Once the relay is closed, the solid-state path is disabled.

3 Analysis and Sizing of the Pre-Charge Dynamics

The pre-charge network is mathematically modeled as a first-order ideal RC circuit. The primary system parameters dictate a DC source voltage (V_{IN}) of 48 V, a total bus capacitance (C_{tot}) of 7920 μF, and a physical current-limiting series resistance (R_{pre}) of 50 Ω.

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Ritaglio della zona di pre-carica dell'LM5069 dal file "Good_Boy_Limiter_escheme.pdf"

Figure 1: Dettaglio della topologia di pre-carica allo stato solido. Il resistore R3 da 50 Ω domina l'impedenza del ramo, limitando fisicamente la corrente massima a 0.96 A indipendentemente dallo stato del controller LM5069.

3.1 Peak In-Rush Current Constraint

The theoretical peak current (I_{peak}) occurs instantaneously at $t = 0$, assuming the capacitor bank is fully discharged ($V_C = 0$ V). The peak current is defined by Ohm's Law across the series resistor:

$$I_{peak} = \frac{V_{IN}}{R_{pre}} = \frac{48 \text{ V}}{50 \ \Omega} = 0.96 \text{ A} \tag{2}$$

By actively restricting the peak current to a mere 0.96 A, the GBL architecture completely eliminates the stress on the upstream 48 V power supply. The upstream source perceives the startup event as a negligible, highly controlled 46 W load rather than a multi-kilowatt short circuit.

3.2 Time Constant and RC Charging Profile

The exponential voltage charging profile of the capacitor bank is governed by the RC time constant (τ), which dictates the temporal response of the circuit:

$$\tau = R_{pre} \times C_{tot} = 50 \ \Omega \times 0.00792 \text{ F} = 0.396 \text{ s} \tag{3}$$

The bus voltage as a function of time, $V_C(t)$, is defined by the standard step-response differential equation solution:

$$V_C(t) = V_{IN} \left(1 - e^{\frac{-t}{\tau}}\right) = 48 \left(1 - e^{\frac{-t}{0.396}}\right) \tag{4}$$

In standard engineering practice, a capacitor is considered functionally charged at 3τ (95.0% of V_{IN}) and fully charged at 5τ (99.3% of V_{IN}).

We can calculate the exact time required to reach the 95% threshold (45.6 V):

$$t_{95\%} = 3 \times \tau = 3 \times 0.396 \text{ s} = 1.188 \text{ s} \tag{5}$$

Furthermore, the time required to reach the 99% threshold (47.5 V):

$$t_{99\%} = 5 \times \tau = 5 \times 0.396 \text{ s} = 1.980 \text{ s} \tag{6}$$

These timing calculations define the minimum operational window required before the ESP32-S3 firmware can safely command the main mechanical relay to close. Actuating the relay before 1.188 s would risk residual arcing due to a significant remaining voltage differential.

3.3 Energy Transfer Thermodynamics

The total energy transferred from the 48 V source to fully charge the capacitor bank is derived from the energy stored in the electric field of the capacitors:

$$E_{cap} = \frac{1}{2} C_{tot} V_{IN}^2 = \frac{1}{2} \times 0.00792 \text{ F} \times (48 \text{ V})^2 = 9.1238 \text{ J} \tag{7}$$

A fundamental principle of thermodynamics in RC charging circuits dictates that an exactly equal amount of energy is dissipated as heat within the resistive elements of the charging path. Therefore, regardless of the charging speed, exactly 9.12 J of thermal energy is deposited into the 50 Ω wire-wound resistor during every startup sequence. This will be critically analyzed in Section 6.

4 In-Rush Controller (LM5069) Analysis and Sizing

The Texas Instruments LM5069-2 operates as the gate driver and primary hardware safety controller for the solid-state pre-charge branch.

The following subsections detail the mathematical sizing of its internal thresholds.

4.1 Undervoltage Lockout (UVLO) Sizing

The UVLO pin ensures the controller only activates when the input bus is stable and possesses sufficient voltage to properly drive the downstream loads. The LM5069 possesses a precise internal comparator with an enable threshold of 2.5 V. To prevent rapid ON/OFF oscillation (chattering) when the input voltage fluctuates near the threshold, the IC utilizes an internal 21 μA current source to provide a programmable hysteresis.

The GBL schematic defines a dedicated UVLO resistive divider utilizing $R_{25} = 143$ kΩ as the top resistor and $R_{30} = 10$ kΩ as the bottom resistor connected to ground.

The rising input threshold voltage ($V_{\text{UVLO\_RISE}}$) at which the controller enables the pre-charge sequence is calculated as:

$$V_{\text{UVLO\_RISE}} = V_{REF} \times \frac{R_{25} + R_{30}}{R_{30}} = 2.5\text{ V} \times \frac{143\text{ k}\Omega + 10\text{ k}\Omega}{10\text{ k}\Omega} = 38.25\text{ V} \tag{8}$$

Once the input voltage exceeds 38.25 V, the 21 μA internal current source activates and flows out of the UVLO pin, through the bottom resistor R_{30}, effectively raising the pin voltage. This creates the hysteresis voltage ($V_{\text{UVLO\_HYS}}$), which is dictated by the current source acting upon the top resistor R_{25}:

$$V_{\text{UVLO\_HYS}} = I_{HYS} \times R_{25} = 21\ \mu\text{A} \times 143\text{ k}\Omega = 3.003\text{ V} \tag{9}$$

The falling threshold voltage ($V_{\text{UVLO\_FALL}}$), where the controller will abort the sequence due to a sagging input bus, is therefore:

$$V_{\text{UVLO\_FALL}} = V_{\text{UVLO\_RISE}} - V_{\text{UVLO\_HYS}} = 38.25\text{ V} - 3.003\text{ V} = 35.247\text{ V} \tag{10}$$

Therefore, the pre-charge sequence is hardware-locked until the input voltage is a steady 38.25 V, and it possesses a robust 3 V noise margin before shutting down at 35.24 V.

4.2 Overvoltage Lockout (OVLO) Sizing

Similarly, the OVLO pin protects the downstream systems from catastrophic input voltage surges, such as load-dump events or incorrect power supply connections. To maintain precise and independent control over the overvoltage threshold, the GBL schematic utilizes a separate, dedicated divider string: $R_{32} = 210$ kΩ as the top resistor and $R_{29} = 10$ kΩ as the bottom resistor.

The rising overvoltage cutoff threshold ($V_{\text{OVLO\_RISE}}$) is calculated using the same 2.5 V internal reference:

$$V_{\text{OVLO\_RISE}} = V_{REF} \times \frac{R_{32} + R_{29}}{R_{29}} = 2.5\text{ V} \times \frac{210\text{ k}\Omega + 10\text{ k}\Omega}{10\text{ k}\Omega} = 55.00\text{ V} \tag{11}$$

The OVLO pin also features a 21 μA hysteresis current source. The hysteresis voltage is:

$$V_{\text{OVLO\_HYS}} = I_{HYS} \times R_{32} = 21\ \mu\text{A} \times 210\text{ k}\Omega = 4.410\text{ V} \tag{12}$$

The falling recovery threshold ($V_{\text{OVLO\_FALL}}$), where the system permits operation to resume after an overvoltage event clears, is:

$$V_{\text{OVLO\_FALL}} = V_{\text{OVLO\_RISE}} - V_{\text{OVLO\_HYS}} = 55.00\text{ V} - 4.41\text{ V} = 50.59\text{ V} \tag{13}$$

The hardware overvoltage cutoff is situated at 55.00 V. This provides adequate operational margin above the nominal 48 V state, while strictly protecting downstream components from transient degradation.

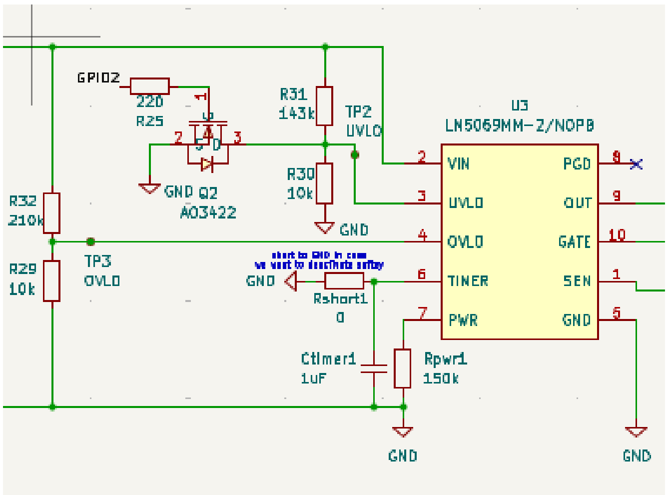

Figure 2: Schematic of the independent resistive dividers for the undervoltage (R_{25}, R_{30}) and overvoltage (R_{32}, R_{29}) thresholds. The use of two separate networks ensures clear and independent hysteresis for the operating windows at 38.25 V and 55.00 V.

4.3 Hardware Current Limit (I_{LIM})

The standard operational mode of the LM5069 relies on sensing the voltage drop across a precision shunt resistor (R_{SENS1}) placed directly between the **VIN** and **SENSE** pins. If the load current increases such that the voltage across the sense resistor reaches 55 mV, the LM5069 limits the current by dynamically adjusting the MOSFET gate voltage.

In the GBL design, a value for $R_{SENS1} = 20\ \text{m}\Omega$ is utilized. The absolute hardware current limit is established mathematically as:

$$I_{LIM} = \frac{V_{SNS(th)}}{R_{SENS1}} = \frac{55\ \text{mV}}{20\ \text{m}\Omega} = 2.75\ \text{A} \tag{14}$$

4.4 Power Limit Sizing ($P_{FET(LIM)}$)

To protect the external MOSFET from violating its Safe Operating Area (SOA) during linear mode regulation, the LM5069 allows the maximum allowable power dissipation of the FET to be programmed via the PWR pin. The LM5069 monitors the V_{DS} of the MOSFET by comparing the SENSE pin and the OUT pin.

This limit is set using an external resistor (R_{PWR1}) connected to the PWR pin. The schematic indicates $R_{PWR1} = 150\ \text{k}\Omega$.

The formula correlating the programmable power limit to the sense resistor and the PWR resistor is provided by Texas Instruments as:

$$P_{FET(LIM)} = \frac{R_{PWR1}}{1.25 \times 10^5 \times R_{SENS1}} = \frac{150,000\ \Omega}{1.25 \times 10^5 \times 0.020\ \Omega} = 60\ \text{W} \tag{15}$$

The hardware is theoretically configured to throttle the MOSFET gate if it detects the MOSFET is dissipating more than 60 W of heat. However, the presence of the 50 Ω resistor ensures the MOSFET never dissipates anywhere near 60 W. This will be proven in Section 5.

4.5 Fault Timer Duration (t_{FAULT})

The fault timeout allows the system to ride through brief, acceptable transient over-currents without nuisance tripping, but ensures the MOSFET is definitively shut down if a fault persists long enough to threaten the system. The timing is dictated by an external capacitor $C_{TIMER1} = 1\ \mu\text{F}$ connected to the TIMER pin.

The typical duration before a fault is declared and the gate is pulled low is defined by the equation:

$$t_{FAULT} = C_{TIMER1} \times 4.7 \times 10^4 = 1 \times 10^{-6}\ \text{F} \times 4.7 \times 10^4 = 0.047\ \text{s} = 47\ \text{ms} \tag{16}$$

The specific LM5069-2 variant utilized in this design features an Auto-Retry mechanism. If the controller were to enter a current or power limit regulation state for longer than the 47 ms timer, the device would shut down the MOSFET, run through an extended internal cool-down cycle (dictated by the same capacitor), and then attempt to restart the process automatically. However, as established, the controller is physically prevented from ever entering the regulation state due to the 50 Ω resistor.

5 Thermal Management and MOSFET Safe Operating Area (SOA)

The defining architectural characteristic of the GBL pre-charge circuit is the intentional shifting of the massive thermal dissipation burden away from the semiconductor lattice of the STB15810 MOSFET and onto the robust, passive wirewound resistor.

5.1 The STB15810 MOSFET in Deep Saturation

In the Good Boy Limiter architecture, the presence of the 50 Ω resistor physically restricts the maximum current to 0.96 A. Because 0.96 A is vastly lower than the LM5069's 2.75 A current regulation threshold, the LM5069's internal high-side charge pump continues to pump unabated. It fully enhances the MOSFET gate, raising V_{GS} to its maximum value (typically 10 V to 12 V above the source). This pushes the MOSFET completely out of the linear region and deep into the ohmic region (saturation).

The STB15810 is a heavily over-specified 100 V, 110 A device utilizing advanced STripFET™ F7 trench gate technology, boasting an ultra-low $R_{DS(on)}$ of 3.9 mΩ typical and 7 mΩ absolute maximum.

Assuming the worst-case scenario at maximum operating junction temperature where $R_{DS(on)} =$ 7 mΩ, the peak power dissipated by the MOSFET during the absolute highest current flow of the pre-charge phase (0.96 A at $t = 0$) is simply calculated using Joule's law for a pure resistance:

$$P_{MOSFET(peak)} = I_{peak}^2 \times R_{DS(on)} = (0.96\ \text{A})^2 \times 0.007\ \Omega = 0.0064\ \text{W} = 6.4\ \text{mW} \tag{17}$$

Dissipating a mere 6.4 mW means the STB15810 will experience absolutely zero measurable thermal rise above ambient temperature. The device remains entirely safe, utterly immune to the Spirito effect, and operates infinitely deep within the safest bounds of its DC SOA curve. From the perspective of thermal stress, the MOSFET acts as a perfect wire.

6 Thermal Limitations: Analyzing the 50 Ω Power Resistor

Because the MOSFET acts as a near-perfect conductor, the 50 Ω wirewound resistor assumes basically 100% of the thermal load during the pre-charge event.

During the exact moment of turn-on ($t = 0$), the capacitor bank acts as a short circuit, meaning the entire 48 V source is dropped across the resistor. The instantaneous peak power dissipation is:

$$P_{RESISTOR(peak)} = \frac{V_{IN}^2}{R_{pre}} = \frac{(48\ \text{V})^2}{50\ \Omega} = 46.08\ \text{W} \tag{18}$$

The selected component, EWWR0010J50R0T9, is a wirewound power resistor rated for 10 W of continuous steady-state power. Operating a 10 W rated resistor at 46.08 W could appears as a design error; however, it is permissible because pre-charging is a highly transient pulse event, not a steady-state condition.

As calculated in Equation (7), the total energy deposited into the resistor over the entire 1.5 to 2.0 second charging curve is strictly bounded to 9.12 J. Wirewound resistors are specifically constructed by winding a resistive alloy wire around a dense ceramic core. This ceramic core possesses a significant thermal mass. During a short 2-second pulse, the heating is largely adiabatic, the energy is absorbed directly into the thermal mass of the ceramic core, resulting in a moderate, entirely safe transient temperature rise.

The critical operational limitation here is duty cycle: the system must not be commanded to repeatedly toggle the pre-charge sequence in rapid succession. Sufficient cool-down time (several seconds) must be enforced by the ESP32 software to allow the ceramic core to shed this thermal energy to the ambient environment.

7 Control Logic, System Security, and the Software Timeout

While the delegation of the current-limiting role to a passive 50 Ω resistor safely protects the MOSFET, it simultaneously creates a critical vulnerability within the hardware protection layers of the LM5069. This vulnerability necessitates the implementation of an active software safety supervisory system via the ESP32-S3 microcontroller.

7.1 The "Hardware Blind Spot" during a Hard Short

Consider a catastrophic operational scenario: a dead short circuit directly to ground exists on the 48 V output bus (e.g., a crushed cable or a failed downstream component) before the system is turned on.

When the ESP32 initiates the pre-charge sequence, the 50 Ω resistor performs its physical duty and limits the fault current:

$$I_{fault} = \frac{48\ \text{V}}{50\ \Omega + 0.020\ \Omega + 0.007\ \Omega} \approx 0.96\ \text{A} \tag{19}$$

Because the fault current (0.96 A) is significantly lower than the LM5069 hardware current limit threshold (2.75 A), the LM5069 internal current comparator is never triggered. Furthermore, because the 50 Ω resistor is dropping nearly the entire 48 V, the voltage at the SENSE pin is roughly 0 V, and the V_{DS} across the MOSFET is negligible (a few millivolts). Consequently, the power limit comparator ($P_{LIM} = 60$ W) is also completely blind to the fault.

The Catastrophic Result: The LM5069 views the dead short circuit as a completely normal, healthy operating condition. The internal 47 ms fault timer (C_{TIMER}) never begins counting. The hardware safety system will remain engaged indefinitely, forcing the 10 W wirewound resistor to continuously dissipate 46.08 W. Within a matter of seconds, the resistor will violently exceed its thermal degradation limits, resulting in extreme overheating, outgassing, potential ignition of surrounding components, and ultimately an open-circuit failure.

7.2 The ESP32-S3 Software Timeout Mitigation

To permanently resolve this dangerous topological vulnerability, the ESP32-S3 microcontroller is tasked with monitoring the dynamic health of the pre-charge process via the dedicated Voltage Sensing Modules.

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Ritaglio dei moduli V_Sensing dal file "Good_Boy_Limiter_escheme.pdf"

Figure 3: Schema elettrico dei moduli V_Sensing utilizzati per la lettura della tensione di bus da parte dell'ESP32. Si notano i partitori resistivi accoppiati ai diodi BAT54S, essenziali per proteggere gli ingressi ADC del microcontrollore da sovratensioni transitorie.

The ESP32 reads the intermediate bus voltage via a resistor divider string ($R_{14} = 10$ kΩ, etc.). These inputs are aggressively protected by dual BAT54S Schottky clamping diodes, which prevent the microcontroller's sensitive ADC pins from exceeding the 3.3 V logic supply limit even during severe transients.

The ESP32 firmware must be programmed to implement a strict, non-blocking state-machine timeout logic:

1. **Initiation:** The ESP32 asserts the LM5069 EN/UVLO pin HIGH via a GPIO pin to begin the pre-charge sequence. Simultaneously, it initializes an internal hardware software timer ($t = 0$).

2. **Continuous Monitoring:** The ADC continuously samples the bus voltage (V_{bus}) at a high frequency.

3. **Success Condition:** If V_{bus} successfully exhibits the standard RC charging curve and exceeds 45 V (approximately 3τ) within a 1.5 to 2.0 second window, the pre-charge is deemed successful. The ESP32 then commands the main 12 V mechanical relay to close, routing the 260 A capability to the load. Once the relay confirms closure, the solid-state branch is deactivated.

4. **Timeout Failure Condition (The Critical Mitigation):** If V_{bus} fails to rise above a minimum safety threshold within 3.0 seconds (a "Timeout of Security"), it is definitive proof of a downstream fault or short circuit absorbing the charge current. The ESP32 must immediately and unconditionally force the EN/UVLO pin LOW. This action forcefully shuts down the LM5069, halting the current flow and saving the 50 Ω resistor from thermal destruction.

This software-defined safety boundary bridges the gap left by the hardware topology, ensuring full system resilience against all modes of failure.

8 Auxiliary Power Supply: LM5013 DC-DC Converter

To drive the heavy inductive coil of the main mechanical relay and provide logic-level power to the digital supervision systems, the GBL integrates a high-voltage auxiliary step-down (buck) converter based on the Texas Instruments LM5013DDAR.

The LM5013 steps the primary 48 V distribution bus down to an intermediate 12 V rail. The converter is capable of handling inputs up to 100 V, providing massive immunity against 48 V bus ringing and transients. It operates utilizing a Constant On-Time (COT) architecture. COT control provides exceptional transient load response (essential when slamming a heavy relay coil) and entirely eliminates the need for complex external loop compensation networks (type II or type III compensators), simplifying the PCB layout.

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Ritaglio del DC-DC LM5013 dal file "Good_Boy_Limiter_escheme.pdf"

Figure 4: Topologia del convertitore Step-down LM5013. Si evidenziano i componenti principali: l'induttore L3 da 22 μH, il diodo di ricircolo Dsw1 (SS510C) e la rete di feedback R23/R24 per la precisa regolazione dell'uscita a 12 V.

8.1 Feedback Sizing for the 12 V Rail

The output voltage is programmed via a precision resistor divider feeding back into the FB pin. The highly precise internal bandgap reference voltage of the LM5013 is exactly $V_{REF} = 1.2$ V. The schematic defines the top feedback resistor as $R_{FB1}(R_{23}) = 453$ kΩ and the bottom resistor connected to ground as $R_{FB2}(R_{24}) = 49.9$ kΩ.

The nominal output voltage is mathematically derived as:

$$V_{OUT} = V_{REF} \times \left(1 + \frac{R_{FB1}}{R_{FB2}}\right) = 1.2 \text{ V} \times \left(1 + \frac{453 \text{ k}\Omega}{49.9 \text{ k}\Omega}\right) \tag{20}$$

$$V_{OUT} = 1.2 \text{ V} \times (1 + 9.078) = 1.2 \text{ V} \times 10.078 = 12.09 \text{ V} \tag{21}$$

This 12.09 V rail is perfectly suited for driving standard automotive and industrial relay coils, which typically specify an operational envelope of 9 V to 14 V.

8.2 Core Buck Components and Filtering

The power stage utilizes a low-DCR inductor $L_3 = 22$ μH and an SS510C Schottky freewheeling diode (D_{SW1}). The presence of the external Schottky diode is mandatory for this non-synchronous topology to provide a path for the inductor current during the MOSFET off-time. The Constant On-Time programming resistor is $R_{ON}(R_{21}) = 100$ kΩ, establishing the nominal switching frequency.

Input voltage ripple and filtering are aggressively handled by dual low-ESR 2.2 μF ceramic capacitors (C_{13}, C_{14}) plus a larger 47 μF bulk capacitor (C_{15}). This network attenuates high-frequency switching noise, preventing the buck converter from injecting harmonic distortions back into the main 48 V distribution bus. A secondary step-down stage (utilizing the SY8120B buck converter) subsequently drops this intermediate 12 V to a highly regulated 3.3 V rail to power the ESP32-S3 microcontroller and its associated sensitive analog peripherals.

9 Conclusion and Recommendations

The "Good Boy Limiter" architecture represents a possible solution to the challenge of managing massive capacitive loads in high-current DC environments. By utilizing a mixed-topology parallel design, it combines the zero-loss continuous conduction capabilities of a mechanical relay with the non-destructive startup characteristics of a solid-state pre-charge circuit.

The sizing of the Texas Instruments LM5069-2 components is mathematically made. The selected resistor dividers establish 38.25 V and 55.00 V operational voltage lockouts, ensuring the system only operates within a safe 48 V nominal window. The absolute hardware current limit is determinate by R_{SENS1} at 2.75 A.

Furthermore, the decision to utilize a 50 Ω series power resistor shields the STB15810 MOSFET from the destructive Safe Operating Area (SOA) stresses typically associated with linear-mode hot-swap events. By pushing the MOSFET deep into saturation, its thermal load is reduced to a negligible 6.4 mW, transferring the energy entirely to the adiabatic mass of the wirewound resistor.

Therefore, it is an absolute, critical project requirement to implement the 3.0 second Software Safety Timeout logic on the ESP32-S3. Without this digital supervisor actively pulling the LM5069 EN pin low upon a failure to charge, the 10 W pre-charge resistor will face inevitable and thermal failure under a continuous 46 W fault load.